

Hybrid resolution refinement and the rodeo algorithm for eigenstate preparation: a demonstration on the pairing Hamiltonian

Scott Bogner,¹ Cecilie Glittum,² Heiko Hergert,¹ Morten Hjorth-Jensen,^{2,*}
Gunnar Lange,² Ryan LaRose,³ Dean Lee,¹ and Francesco Pietro Massel^{4,2}

¹*Facility for Rare Isotope Beams and Department of Physics and Astronomy,
Michigan State University, East Lansing, MI 48824, USA*

²*Department of Physics, University of Oslo, N-0316 Oslo, Norway*

³*Department of Computational Mathematics, Science and Engineering and Facility for Rare Isotope Beams,
Michigan State University, East Lansing, MI 48824, USA*

⁴*Department of Science and Industry Systems, University of South-Eastern Norway, N-3616 Kongsberg, Norway*
(Dated: July 20, 2026)

Projective spectral filters such as the rodeo algorithm prepare eigenstates with exponential accuracy, but only from an input state with sizable overlap p with the target — the quantity that vanishes at scale for variational and adiabatic preparation alone. We combine *resolution refinement*, which bootstraps an eigenstate from a coarse to a fine model space along a gapped adiabatic path, with the rodeo algorithm, and demonstrate the hybrid end to end on the exactly solvable constant-pairing Hamiltonian. For four particles on four levels (eight qubits) the refined input ($p \simeq 0.998$) is filtered to infidelity $\sim 10^{-6}$ within one or two rodeo cycles, and the extracted adiabatic time scales linearly with the inverse spectral gap along the path across weak to strong pairing. Every stage is compiled to explicit rotation-CNOT circuits, and the complete pipeline — Trotterized refinement and Trotterized controlled evolution included — is simulated at the circuit level, confirming that modest product-formula depths do not spoil the filter. A transparent gate budget compares the optimizer-free pipeline with a full-space UCCSD-VQE preparation, whose accuracy floor the pipeline undercuts by one to three orders of magnitude depending on the Trotter budget. The construction provides a concrete template for eigenstate preparation in nuclear-structure applications on mid-circuit-measurement hardware.

I. INTRODUCTION

Determining the low-lying eigenvalues and eigenvectors of a many-body Hamiltonian is one of the grand challenges of quantum many-body theory, where the Hilbert-space dimension grows exponentially with the number of constituents. Quantum computers offer new routes to eigenvalue estimation and eigenstate preparation, and their prospects for nuclear structure have been mapped in recent community studies [8, 36, 37], with schematic nuclear models — pairing, Lipkin, and shell-model Hamiltonians — serving as the standard proving ground [22–24, 26–29]. The state-of-the-art preparation methods, however, each face an obstacle at scale. The variational quantum eigensolver (VQE) [1] is limited by the cost of its classical optimization loop, by the accuracy floor of its ansatz and, for sufficiently expressive circuits, by barren plateaus — gradients that vanish exponentially with system size [2, 35]. Adiabatic state preparation requires evolution times that diverge if the gap between the tracked state and the rest of the spectrum closes along the path [3]. Projective spectral filters — quantum phase estimation [4], polynomial filters based on quantum signal processing [33, 34], projection by measurement [25], and the rodeo algorithm (RA) [5, 6, 30–32] — converge exponentially in the number of cycles, but only once the initial state has a significant squared overlap p with the

target eigenstate; their cost grows as $1/p$, and p itself typically falls exponentially with system size.

These limitations are complementary, which suggests a hybrid strategy. *Resolution refinement* [7] bootstraps an eigenstate across model-space scales: an eigenstate of a coarse (low-resolution) Hamiltonian is lifted by a prolongation operator into a finer space and adiabatically evolved to the fine-resolution eigenstate. Because refinement preserves the low-energy spectral structure, the path stays gapped and the required adiabatic time scales only as the inverse physical gap (times a slowly growing system-size factor), delivering an output with $p \approx 1$. That output is exactly the high-overlap input the rodeo algorithm needs to converge exponentially. This is the core idea of the present approach: (i) prepare a low-resolution eigenstate; (ii) prolong and adiabatically refine to a final time T^* ; (iii) apply the rodeo algorithm to reach the target fidelity and observables with exponential convergence.

In this work we realize the full pipeline end to end on a transparent and exactly solvable testbed, the constant-pairing Hamiltonian familiar from nuclear and condensed-matter many-body theory [17] and integrable by the Richardson ansatz [19–21]. Its exact solvability is precisely what makes it the right laboratory: every stage of the pipeline can be validated quantitatively — the gate-level UCCSD coarse solver against full configuration interaction (FCI) and coupled-cluster doubles (CCD); the prolongation and adiabatic refinement against the exact high-resolution eigenstate; and

* mhjensen@uio.no

the rodeo filter against the analytic success-probability formulas of Ref. [6]. All circuits are implemented explicitly as rotation and CNOT gates (and an ancilla-based rodeo cycle), so the demonstration doubles as a reference implementation for hardware.

Relative to the resolution-refinement proposal of Ref. [7] and the rodeo literature [5, 6], the present work contributes four elements: (i) the first *combination* of the two algorithms, closing the overlap loophole of projective filters with an input whose overlap the refinement itself makes large; (ii) the first application of refinement to a hierarchy of *single-particle bases* in a fermionic pairing problem, where the prolongation is gate-free; (iii) a com-

plete gate-level compilation of every stage — including Trotterized refinement and Trotterized controlled evolution — with a transparent two-qubit-gate budget compared, per accepted preparation and at matched accuracy, against a full-space variational preparation; and (iv) a reproducible reference implementation in which every figure and number derives from a single executable notebook.

II. THE PAIRING MODEL AND ITS QUBIT ENCODING

We study the constant-pairing Hamiltonian

$$H = \delta \sum_{p=0}^{k-1} p \sum_{\sigma=\uparrow\downarrow} n_{p\sigma} - \frac{g}{2} \sum_{p,q=0}^{k-1} a_{p\uparrow}^\dagger a_{p\downarrow}^\dagger a_{q\downarrow} a_{q\uparrow}, \quad (1)$$

with k doubly degenerate single-particle levels of spacing δ (set to $\delta = 1$) and a pairing strength g . The interaction scatters time-reversed pairs between levels; it conserves the seniority (the number of unpaired particles), so the ground state of N paired particles lives entirely in the seniority-zero subspace, and the model is exactly solvable by the Richardson ansatz [19–21] — the property that anchors every benchmark below. We map level p and spin σ to qubit $2p + \sigma$ (so a k -level problem uses $2k$ qubits) and encode the fermionic operators through the Jordan–Wigner transformation. The Hartree–Fock (HF) reference fills the lowest $N/2$ levels, with $E_{\text{HF}}(N=4, g) = 2 - g$.

Throughout we use $N = 4$ particles as the running example. The fixed- N sector dimension is $\binom{2k}{N}$, growing from 1 at $k = 2$ (a single filled determinant) to 70 at $k = 4$ and 1820 at $k = 8$; the seniority-zero ground-state problem is correspondingly compact, which makes the model an ideal multi-resolution laboratory. “Basis refinement” here means *increasing k at fixed N* : the number of single-particle levels (and qubits) grows while the particle number is held fixed.

III. THE HYBRID PIPELINE

A. Coarse-state preparation: gate-level UCCSD–VQE

The coarse eigenstate is prepared with a unitary coupled-cluster ansatz restricted to singles and doubles (UCCSD) [9, 13], Trotterized into a product of excitation circuits acting on the HF reference,

$$|\Psi(\theta)\rangle = \prod_{\mu} e^{\theta_{\mu}(\hat{\tau}_{\mu} - \hat{\tau}_{\mu}^{\dagger})} |\text{HF}\rangle. \quad (2)$$

Through the Jordan–Wigner map each anti-Hermitian generator becomes a sum of commuting Pauli strings, so every factor compiles into a basis change, a CNOT ladder, a single $R_z(\theta)$ rotation, and the uncompute; a single excitation yields two Pauli-rotation sub-circuits and a double excitation yields eight (the circuits are shown in Fig. 9). We optimize with the exact analytic gradient of the product ansatz (the naive two-term parameter shift is incorrect beyond the first factor, because later factors acquire frequency-one cross terms). Because the pairing force does not couple to seniority-changing single excitations, all singles amplitudes vanish to machine precision and the solver is effectively a *doubles* theory, mirroring the classical CCD structure.

B. The prolongation operator

The prolongation P embeds a low-resolution state into the high-resolution space. For basis refinement it maps every many-body basis state of the k_{low} space to the *same* occupation pattern in the k_{high} space, leaving the new single-particle levels empty. In canonical Jordan–Wigner ordering this is gate-free; for lattice refinement the analogous operator distributes each occupied coarse site over the neighboring fine sites with two single-qubit rotations and two CNOTs per site per spatial direction, i.e. $O(N_{\text{sites}})$ gates per dimension [7]. The prolonged state $|\Phi_0\rangle = P|\psi_{\text{low}}\rangle$ satisfies $\langle\Phi_0|H_{\text{high}}|\Phi_0\rangle = E_{\text{low}}$ exactly, and already provides a high-quality first guess for the high-resolution ground state.

C. Resolution refinement (adiabatic phase)

The prolonged state is refined by adiabatically following the interpolated, energy-shifted Hamiltonian

$$H(s) = \cos^2\left(\frac{\pi s}{2}\right) P(H_{\text{low}} - \mu) P^\dagger + \sin^2\left(\frac{\pi s}{2}\right) (H_{\text{high}} - \mu), \quad s = \frac{t}{T}, \quad (3)$$

evolved by a Suzuki–Trotter product over total time $T \leq T^*$. The shift μ (here $\mu = E_{\text{low}} + 0.6$) places the tracked low-energy states below the large null space of $P^\dagger$, so the level followed from $s = 0$ to $s = 1$ is the high-resolution ground state. The instantaneous spectrum stays smooth and gapped along the entire path; consequently the adiabatic time scales as $T \sim 1/\Delta E$ rather than the $O(1/\Delta E_{\text{min}}^2)$ of generic adiabatic preparation, and the overlap $p = |\langle \Psi_{\text{high}} | \Phi(T) \rangle|^2$ rises toward unity.

D. The rodeo algorithm

The rodeo algorithm [5, 6] removes the residual error left by the adiabatic tail. One cycle acts on the system register plus a single ancilla (reused across cycles by mid-circuit measurement; the circuit is shown in Fig. 10): a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} of the system, a phase gate $P(Et_k)$ on the ancilla (which effectively shifts $H \rightarrow H - E$), a second Hadamard, and an ancilla measurement. Post-selecting the ancilla in $|0\rangle$ applies the non-unitary filter $\frac{1}{2}(\mathbb{I} + e^{iEt_k} e^{-iHt_k})$ to the system: an eigenstate $|E_j\rangle$ acquires the factor $e^{i(E-E_j)t_k/2} \cos[(E_j - E)t_k/2]$. For an eigenstate input the probability that all M ancilla measurements return 0 is therefore [6]

$$P_{0M}(E | \{t_k\}) = \prod_{k=1}^M \cos^2\left[\frac{t_k}{2}(E_{\text{obj}} - E)\right], \quad (4)$$

and averaging the t_k over a Gaussian of RMS width σ gives

$$P_{0M}(E) = \left[\frac{1 + e^{-(E_{\text{obj}} - E)^2 \sigma^2 / 2}}{2} \right]^M. \quad (5)$$

On resonance ($E = E_{\text{obj}}$) the success probability is unity at every cycle, while off-resonant weight is suppressed geometrically toward the background 2^{-M} (the arithmetic mean of $\cos^2$ is $1/2$). Scanning E thus locates eigenvalues as peaks whose width $\sim 1/\sigma$ sets the energy resolution; fixing E at the target eigenvalue turns the algorithm into a state-preparation filter.

The controlled evolution is the only system-size-dependent ingredient. On hardware e^{-iHt} is realized by a

Suzuki–Trotter decomposition over the $2k$ qubits [14–16]; for the single-qubit benchmark of Ref. [6] it reduces to a Z – Y – Z Euler rotation $R_{\hat{n}}(\theta) = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2}$ implemented through the Qiskit native gate. In Secs. IV A–IV E we apply the controlled evolution exactly within the conserved N -particle sector to isolate the algorithmic behavior; the circuit-level pipeline with explicit product formulas, and the Trotter error it introduces, are quantified in Sec. V B. Throughout we draw the cycle times t_k from a Gaussian as in Ref. [5]; optimized deterministic [30] or geometric [32] time schedules suppress off-resonant weight faster at fixed runtime and would only improve the numbers reported here, as would the controlled-reversal-gate compilation of the controlled evolution of Ref. [31].

a. Convergence criterion. With E tuned to the target eigenvalue E_0 , the all-zero acceptance probability of an input $|\psi\rangle = \sum_k c_k |E_k\rangle$ after M cycles is

$$P_M = p + \sum_{k \neq 0} |c_k|^2 \left[\frac{1}{2} (1 + e^{-(E_k - E_0)^2 \sigma^2 / 2}) \right]^M, \quad p = |c_0|^2. \quad (6)$$

The target weight p is preserved at every round while off-resonant components decay geometrically, so $P_M \rightarrow p$ and the post-selected fidelity obeys

$$\mathcal{F}_M = \frac{p}{P_M} \simeq \left[1 + \frac{1-p}{p} 2^{-M} \right]^{-1}, \quad 1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}. \quad (7)$$

Reaching fidelity $1 - \delta$ therefore needs only $M \gtrsim \log_2[(1-p)/(p\delta)]$ cycles, and since the expected number of repetitions to obtain an accepted run is $\sim 1/P_M \rightarrow 1/p$, the total controlled-evolution effort scales as $O(|\log \delta|/(p\epsilon))$ at energy resolution ϵ [5]. This is efficient precisely when resolution refinement supplies $p \approx 1$.

IV. RESULTS

A. Coarse solver: UCCSD versus exact references

Table I lists the ground-state energies at $g = 1$ for $N = 4$ as the basis grows from $k = 2$ to $k = 8$. The gate-level UCCSD circuit reproduces the operator-level (statevector) UCCSD to machine precision and stays variational ($E_{\text{UCCSD}} \geq E_{\text{FCI}}$), tracking FCI to $\sim 10^{-3}$ – 10^{-2} , whereas CCD over-binds. Figure 1 compares all solvers (exact, CCD, statevector and gate-level UCCSD, and a

TABLE I. Ground-state energies at $g = 1$, $N = 4$, for the constant-pairing Hamiltonian. FCI is the exact N -sector result; CCD is coupled-cluster doubles; UCCSD is the gate-level circuit of Sec. III A. “#CNOT” is the two-qubit gate count of the UCCSD state preparation.

k	qubits	FCI	CCD	UCCSD	#CNOT
2	4	1.000000	1.000000	1.000000	0
3	6	0.794697	0.794697	0.794697	272
4	8	0.635548	0.630443	0.636987	1312
6	12	0.372664	0.350058	0.378719	—
8	16	0.145559	0.097263	0.157856	—

hardware-efficient ansatz [10]) as a function of k . The hardware-efficient ansatz is exact at $k = 2$ and reaches FCI at $k = 3$ only after a multi-start optimization, and struggles at $k = 4$ — the symmetry-blind landscape that motivates the physically structured UCCSD circuit. Figure 2 shows that the same UCCSD solver remains a tight variational bound all the way to $k = 8$ (sixteen qubits), at the cost of a CNOT count that grows steeply (1312 CNOTs already at $k = 4$); the compiled circuits and their depths are analyzed in Sec. V.

B. Resolution refinement: overlap and energy

At $N = 4$ the low-resolution ($k = 2$) space is a single filled determinant, so the prolonged state is the high-resolution HF reference, with first-guess overlaps 0.917 for refinement to $k = 3$ and 0.868 to $k = 4$. Adiabatic evolution along the gapped path of Eq. (3) drives both quantities toward their exact targets: the overlap rises to $\simeq 0.999$ and the energy falls from $E_{\text{HF}} = 1$ to the FCI value from above — 0.7964 for $k = 3$ (FCI 0.794697) and 0.6397 for $k = 4$ (FCI 0.635548) at $T = 30$ (Fig. 4). The instantaneous spectrum stays open along the whole path (Fig. 3, right), confirming the favorable $T \sim 1/\Delta E$ timescale; the larger $k = 4$ space starts lower in overlap and converges more slowly, exactly as expected from the system-size dependence of the refinement.

C. A nontrivial coarse stage: the $k = 3 \rightarrow 4$ chain

At $N = 4$ the $k = 2$ sector is a single determinant, so in the chain above the coarse “solver” is gate-free and the pipeline’s first stage does no work. To exercise the pipeline with a working coarse stage we start instead from the gate-level UCCSD state at $k = 3$ — a genuine 272-CNOT variational preparation, indistinguishable from FCI at this resolution (Table I) — prolong $3 \rightarrow 4$, and refine along the same path, with the shift μ now referenced to the *variational* coarse energy $\langle \psi_{\text{low}} | H_{\text{low}} | \psi_{\text{low}} \rangle$, the only quantity a device can know. Figure 5 compares the two chains. The better coarse

start raises the prolonged overlap from 0.868 to 0.964 and reaches the target overlap 0.99 at $T^* = 9$ instead of $T^* = 16$; at fixed Trotter-step density the saved refinement time is worth $\sim 2 \times 10^4$ CNOTs, vastly more than the 272 spent on the coarse circuit. Starting higher in the resolution hierarchy is therefore not merely allowed by the pipeline — it is the economical choice, and the trade-off is quantified entirely in two-qubit gates.

D. Coupling scan and the path gap

All results above sit at $g = 1$; Fig. 6 probes the refinement across couplings and tests the $T \sim 1/\Delta E$ claim directly. Scanning $g = 0.3\text{--}4$ at fixed $\mu = E_{\text{low}} + 0.6$ (left panel), the minimum instantaneous gap along the path stays open and nearly constant — it is set by the shift itself, $\Delta E_{\text{min}} \simeq \mu - E_{\text{low}}$, dropping only to 0.57 at $g = 4$ — while the time to reach overlap 0.99 grows from $T^* = 1$ at $g = 0.3$ to $T^* = 65$ at $g = 4$. The growth tracks the initial overlap of the prolonged state, which falls from 0.99 to 0.38 over the same range: at fixed gap, a poorer starting state simply has more rearrangement to push through the same bottleneck. This identifies the practical limiter of the method at strong coupling — not a closing gap but a degrading prolongation — and it is precisely the regime where a coarse start higher in the hierarchy (Sec. IV C) pays off most.

Because the path gap is controlled by the shift, varying μ at fixed $g = 1$ isolates the gap dependence cleanly (right panel): the extracted T^* is linear in the inverse minimum gap, $T^* \approx 17/\Delta E_{\text{min}}$, the behavior expected of a gapped, first-order-in- $1/\Delta E$ refinement path and markedly better than the $1/\Delta E_{\text{min}}^2$ of generic adiabatic preparation. The same scan doubles as the sensitivity study for the shift: the choice $\mu_{\text{buf}} = 0.6$ used throughout is conservative, and enlarging the buffer to 1.5 *shortens* the refinement to $T^* = 6$. (For coarse spaces with more than one state the buffer has a ceiling: the shifted coarse spectrum must stay below the null space of $P^\dagger$.)

E. Rodeo filtering of the refined state

We verified the rodeo primitive in two ways: the explicit ancilla circuit agrees with the compact filter $\frac{1}{2}(\mathcal{K} + e^{iEt}e^{-iHt})$ to 10^{-16} , and the all-zero probabilities reproduce Eqs. (4)–(5), with the Gaussian-averaged single-cycle factor matching $\frac{1}{2}(1 + e^{-(\Delta E \sigma)^2/2})$ to three digits.

The state fed to the filter is the resolution-refined $k = 4$ state of Sec. IV B, with overlap $p = 0.998$ with the exact ground state; for contrast we also use the unrefined prolonged guess Φ_0 ($p = 0.868$). Scanning the target energy E (Fig. 7) produces a peak at the ground-state energy; a coarse scan ($\sigma = 2$) resolves it from the excited band, and a fine scan ($\sigma = 8$) locates it at $E = 0.6352$ against the exact $E_0 = 0.635548$. Starting from the refined state,

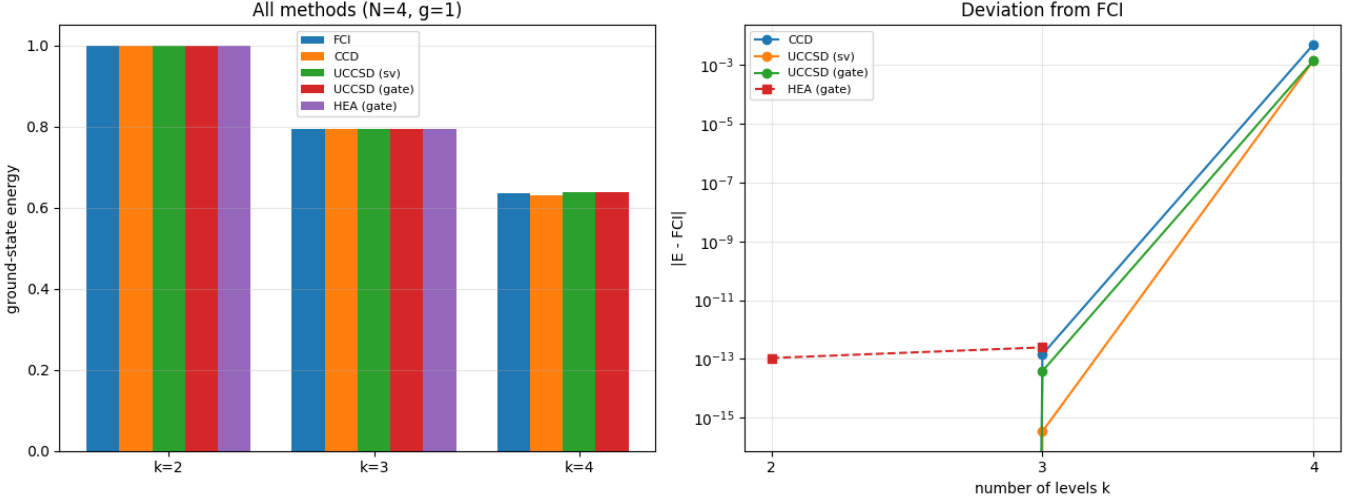


FIG. 1. Ground-state energy at $N = 4$, $g = 1$ obtained with all solvers (left) and their deviation from FCI (right), versus the number of levels k . The structured UCCSD circuits track FCI from above; CCD over-binds; the hardware-efficient ansatz matches FCI for $k \leq 3$ but loses accuracy at $k = 4$.

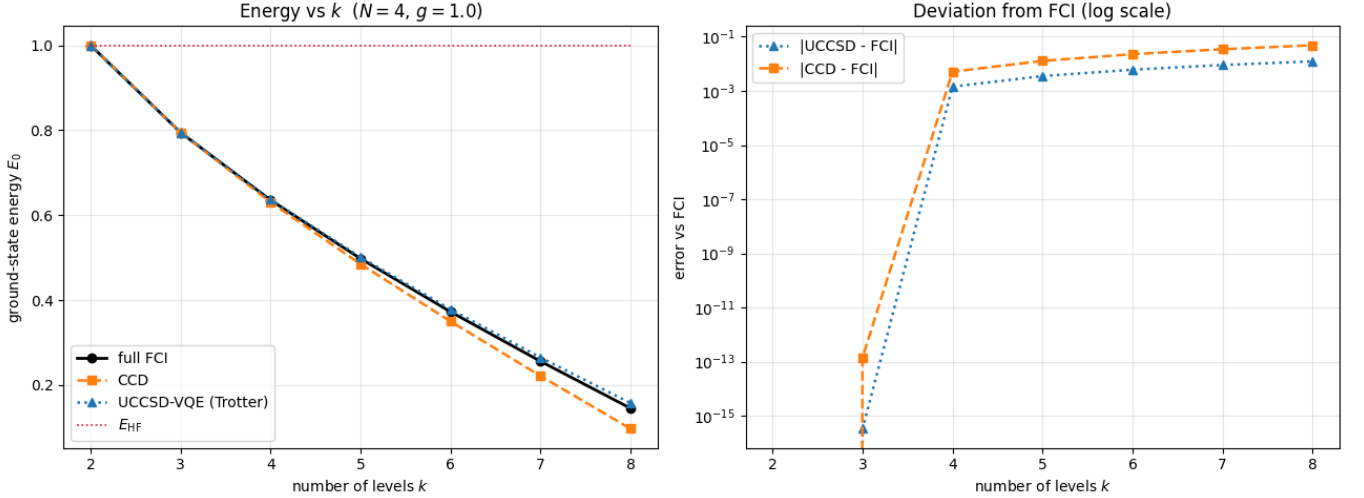


FIG. 2. Scalability of the Trotterized UCCSD-VQE: ground-state energy (left) and deviation from FCI (right) from $k = 2$ to $k = 8$ ($N = 4$, $g = 1$). The same solver remains variational and tight as the basis — and hence the qubit count — grows.

essentially all spectral weight sits in the single ground-state peak; starting from Φ_0 , the small residual weight on excited states is visible as secondary structure above the 2^{-M} background.

Fixing $E = E_0$ and applying M cycles (Fig. 8), the post-selected fidelity rises exponentially and the cumulative acceptance settles at $\approx p$, in quantitative agreement with Eqs. (6)–(7). The refined input reaches an infidelity $1 - \mathcal{F} \simeq 5 \times 10^{-6}$ within one to two cycles; even the cruder prolonged input reaches $\simeq 1.3 \times 10^{-4}$ in a few cycles, while its acceptance holds at 0.868. The dotted reference lines $\frac{1-p}{p}2^{-M}$ bracket the measured infidelities, confirming the convergence criterion. This is the concrete pay-off of the pipeline: resolution refinement supplies the

high overlap that turns the rodeo algorithm from a $1/p$ -expensive filter into a few-cycle polish.

V. QUANTUM CIRCUITS AND DEPTH COMPARISON

All quantum operations in this work are compiled to a common gate set of single-qubit rotations and CNOTs, so the resource costs of the two routes to the high-resolution ground state — a traditional full-space UCCSD-VQE versus the rodeo-terminated pipeline — can be compared gate by gate. Every number quoted below is produced (and verified against exact matrix exponentials) in the accompanying notebook.

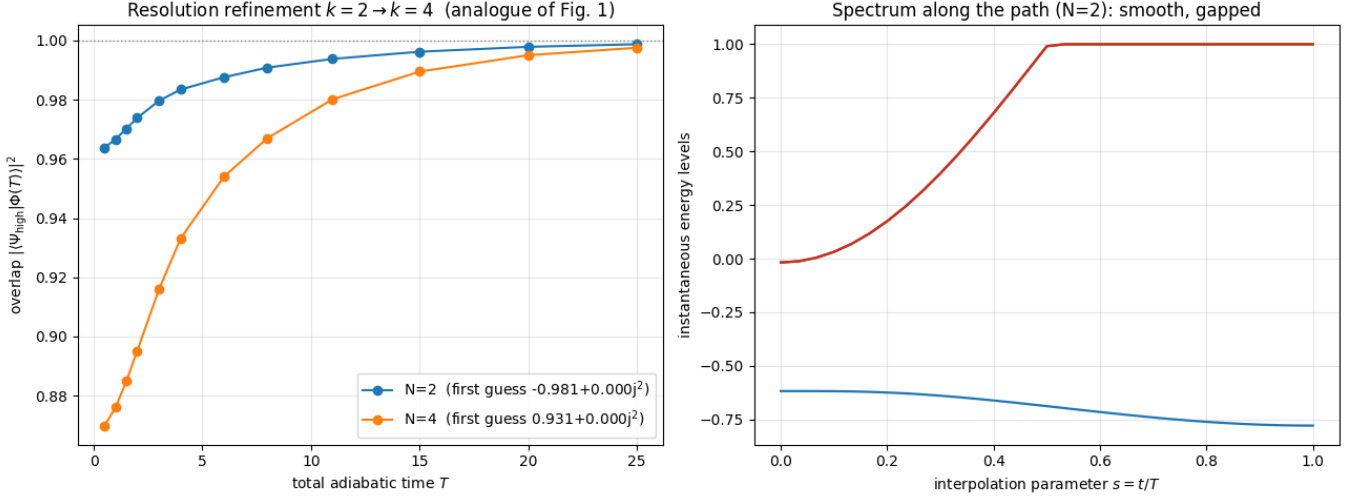


FIG. 3. Resolution refinement $k = 2 \rightarrow k = 4$. Left: overlap of the evolved state with the exact high-resolution ground state versus the total adiabatic time T , for $N = 2$ and $N = 4$. Right: the instantaneous spectrum along the interpolation parameter $s = t/T$ stays smooth and gapped, the condition for efficient refinement.

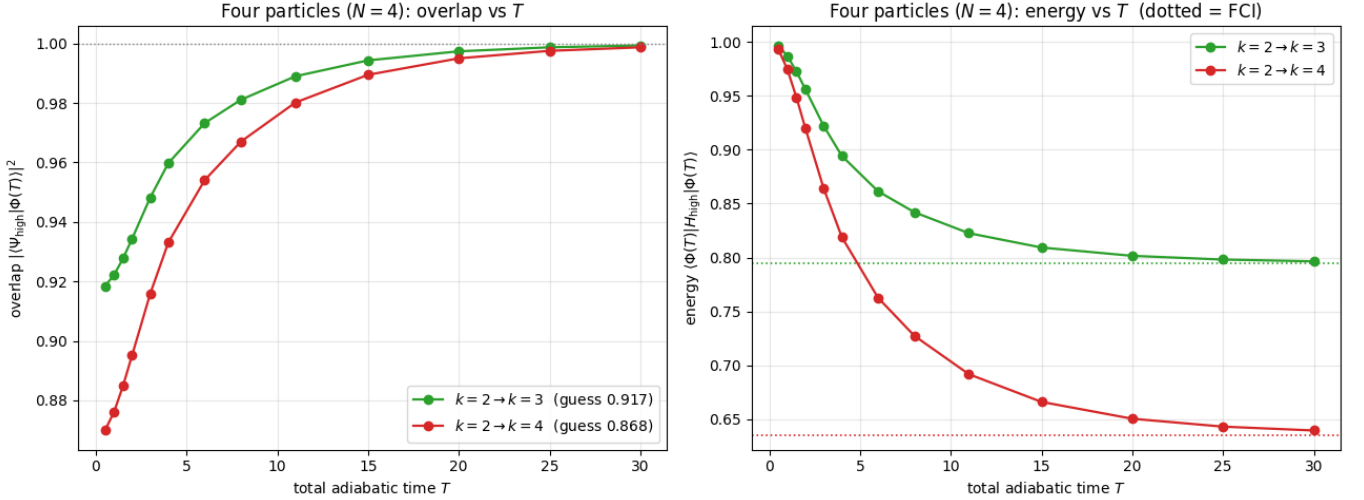


FIG. 4. Four-particle refinement up to $k = 4$: overlap (left) and physical energy $\langle \Phi(T) | H_{\text{high}} | \Phi(T) \rangle$ (right) versus T for $k_{\text{high}} = 3, 4$. The energy approaches the FCI value (dotted) from above while the overlap approaches unity, both on a $\sim 1/\Delta E$ timescale.

A. Circuit building blocks

The elementary primitive is the Pauli rotation $e^{-i\frac{\phi}{2}P}$ for a Pauli string $P = \bigotimes_q \sigma_q$ [18]: each active qubit is rotated into the Z eigenbasis (H for X , $HS^\dagger$ for Y), a CNOT ladder accumulates the parity onto the last active qubit, a single $R_z(\phi)$ is applied, and the ladder and basis changes are uncomputed. A string of weight w therefore costs $2(w-1)$ CNOTs and one parametrized rotation; the circuit reproduces the exact exponential to 3×10^{-16} .

a. UCCSD preparation circuit. Through the Jordan–Wigner map a single excitation compiles into 2 Pauli rotations whose weight grows with the distance

between the spin orbitals (8 CNOTs for adjacent levels, 16 for the $0 \rightarrow 4$ single at $k = 3$), and a generic double excitation into 8 rotations (48–64 CNOTs). The doubles that dominate the pairing problem are special: for a time-reversed pair moving between levels p and q the operators act on the adjacent qubits $(2p, 2p+1)$ and $(2q, 2q+1)$, so the internal Jordan–Wigner Z strings cancel and all 8 Pauli strings have weight 4 — 48 CNOTs per pair-hopping term *independent of the level separation*. The full state preparation at $N = 4$, $g = 1$ then costs 8 excitation operators, 40 rotations and 272 CNOTs at $k = 3$, and 26 operators, 160 rotations and 1312 CNOTs at $k = 4$ (Table I); at $k = 2$ the ansatz is empty (the sector is a single determinant) and the

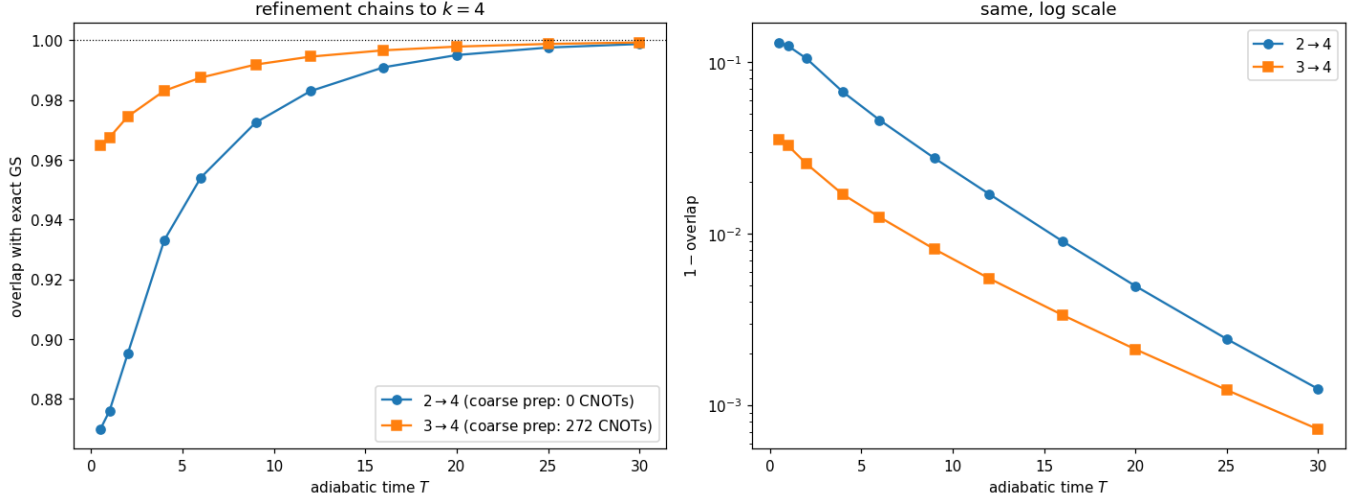


FIG. 5. Refinement chains to $k=4$ at $N=4$, $g=1$, starting from the gate-free $k=2$ determinant (circles) and from the 272-CNOT gate-level UCCSD state at $k=3$ (squares). Left: overlap with the exact ground state versus adiabatic time T ; right: the same on a logarithmic scale. The $3 \rightarrow 4$ chain starts at overlap 0.964 (versus 0.868) and reaches 0.99 in roughly half the adiabatic time.

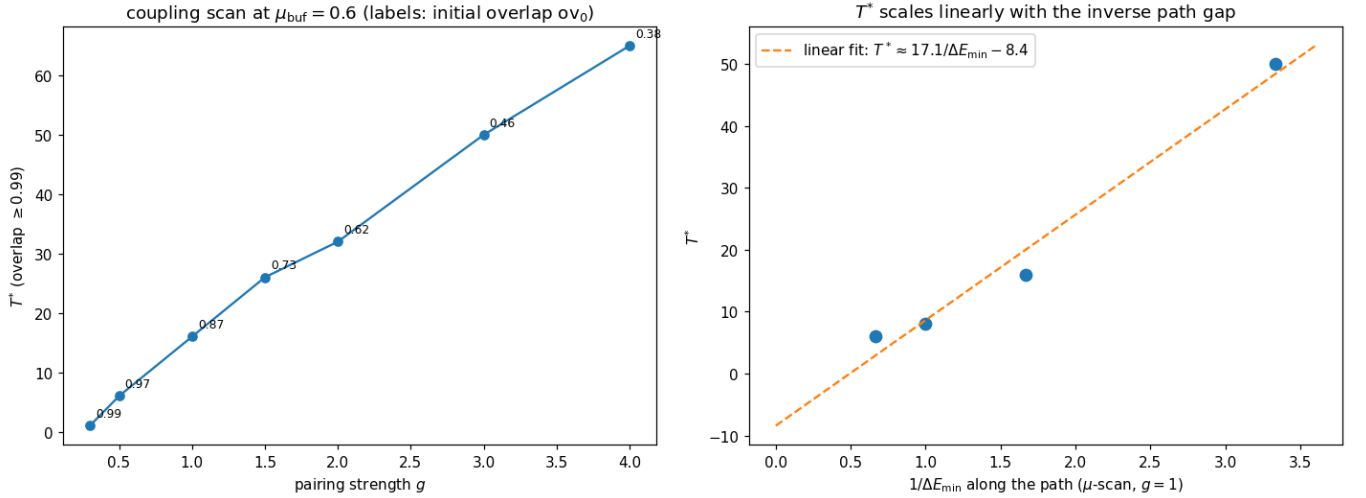


FIG. 6. Left: adiabatic time T^* to reach overlap 0.99 for the $k=2 \rightarrow 4$ refinement versus pairing strength g (point labels: initial overlap of the prolonged state). The minimum path gap stays $\simeq 0.6$ throughout; T^* grows because the starting overlap degrades. Right: T^* versus the inverse minimum path gap, varied through the shift buffer $\mu_{\text{buf}} = 0.3\text{--}1.5$ at $g=1$; the linear fit confirms $T^* \sim 1/\Delta E_{\text{min}}$.

preparation is gate-free.

Figure 9 shows both levels of this hierarchy. Panel (a) is the VQE at the ansatz level: the product of excitation unitaries acts on the HF register, the energy is estimated from repeated measurements, and a classical optimizer updates the P amplitudes θ before the circuit is run again — it is this feedback loop, wrapped around the full-depth state preparation, that multiplies the 1312-CNOT cost into the total gate budget of the traditional route. Panel (b) is the compiled interior of a single weight-4 Pauli rotation, the sub-block that occurs eight times in every pair-hopping double and 160 times

in the $k=4$ preparation; its only parametrized element is the central $R_z(\phi)$, with ϕ fixed by the amplitude of the parent excitation operator.

b. Hardware-efficient ansatz. The HEA [10] used for contrast consists of L layers of R_y rotations on all $2k$ qubits interleaved with a CNOT entangler (nearest-neighbor pairs plus a wrap-around bond), i.e. $2k(L+1)$ variational angles and only $O(2kL)$ CNOTs: 12 angles at $k=2$ ($L=2$) and 30 at $k=3$ ($L=4$). Its depth is far smaller than UCCSD, but the circuit carries no particle-number or seniority structure; a number-penalty term and multi-start optimization are needed at $k=3$, and

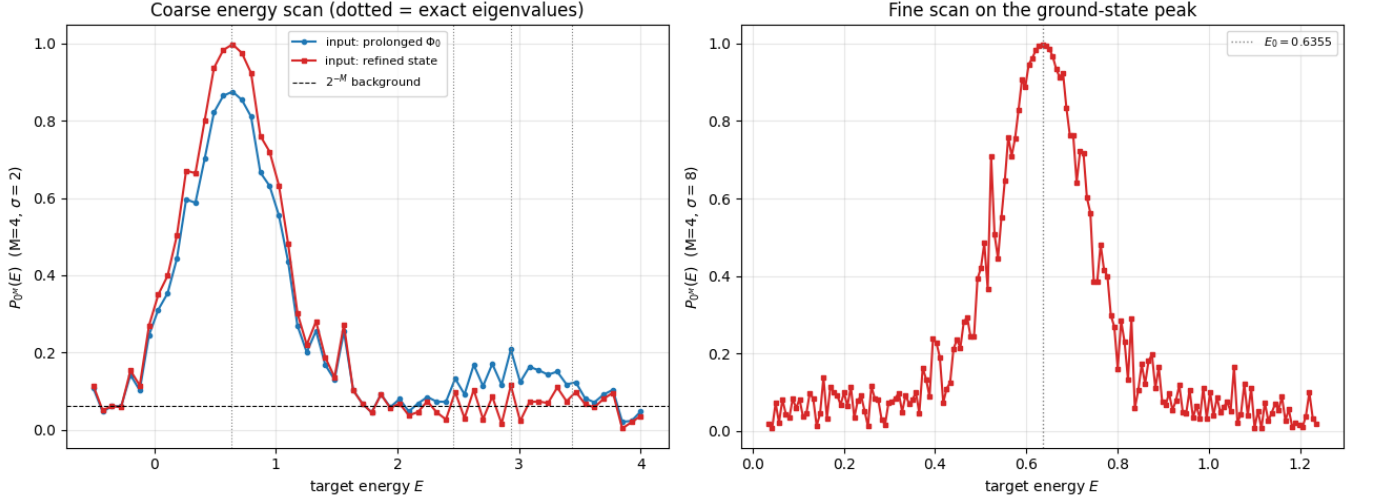


FIG. 7. Rodeo energy scan ($M = 4$). Left: coarse scan ($\sigma = 2$); the dotted vertical lines mark exact eigenvalues and the dashed line the 2^{-M} background. Right: fine scan ($\sigma = 8$) on the ground-state peak, located at $E = 0.6352$ (exact $E_0 = 0.6355$).

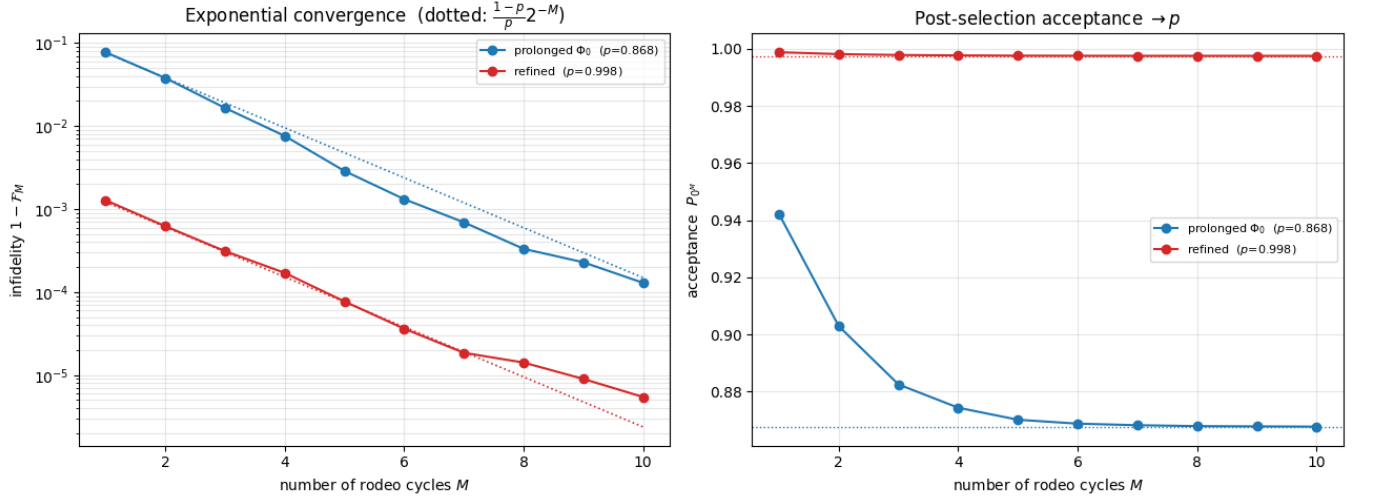


FIG. 8. Rodeo eigenstate preparation at $E = E_0$, averaged over random time sets. Left: post-selected infidelity $1 - \mathcal{F}_M$ versus the number of cycles M (dotted: the bound $\frac{1-\rho}{p}2^{-M}$). Right: the post-selection acceptance converges to the input overlap p .

at $k = 4$ the optimizer fails to reach the ground state (Fig. 1). Depth is not the only currency: trainability of the symmetry-blind landscape becomes the bottleneck first.

c. Rodeo cycle. One rodeo cycle (Fig. 10) adds a Hadamard on the ancilla, the controlled evolution e^{-iHt_k} , the ancilla phase $P(Et_k)$ — the element that effectively shifts $H \rightarrow H - E$ — a second Hadamard, and a mid-circuit measurement; the single ancilla is reset and reused across the M cycles. The contrast with Fig. 9(a) is the structural point of this section: the rodeo circuit contains *no* variational parameters and closes no optimization loop. The times t_k are drawn once from a Gaussian, the target energy E is a classical setting, and the algorithm is driven entirely by the measurement record

— a run is accepted when all M ancilla outcomes read 0, which applies the filter of Sec. IIID cycle by cycle. Where the VQE re-executes its full-depth circuit to steer θ , the rodeo circuit is executed as printed, and its only system-size-dependent element is the controlled evolution. The controlled evolution is compiled from the same Pauli-rotation primitive: in each rotation sub-circuit only the central R_z needs to be controlled (if the rotation is absent the basis changes and CNOT ladder cancel), and a controlled R_z costs 2 extra CNOTs. For the pairing Hamiltonian (1) a first-order Trotter step contains the $k(k-1)/2$ pair-hopping terms (8 weight-4 rotations, 48 CNOTs each), k diagonal ZZ rotations (2 CNOTs each) and $2k$ single-qubit phases, i.e.

$$N_{\text{CNOT}}^{\text{step}}(k) = 24k(k-1) + 2k \quad (8)$$

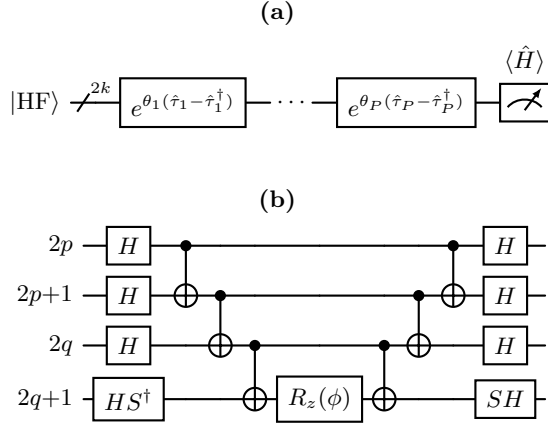


FIG. 9. The variational quantum eigensolver as a circuit. (a) Ansatz level: the Trotterized UCCSD product acts on the HF register ($2k$ qubits, slashed wire) and the energy is estimated by measuring $\langle \hat{H} \rangle$ (in practice a sum of Pauli expectation values, each averaged over many shots); a classical optimizer updates θ and the entire circuit is re-executed, once per energy or gradient evaluation. (b) Compiled circuit of one weight-4 Pauli rotation, here $e^{-i\frac{\phi}{2}X \otimes X \otimes X \otimes Y}$ acting on the four qubits of a pair-hopping double excitation $(2p, 2p+1) \rightarrow (2q, 2q+1)$: basis changes into the Z eigenbasis (H for X , $HS^\dagger$ for Y), a CNOT parity ladder, the single parametrized $R_z(\phi)$, and the uncompute. Eight such blocks (48 CNOTs) build one double excitation; 160 of them (1312 CNOTs) build the $k = 4$ preparation of Table I.

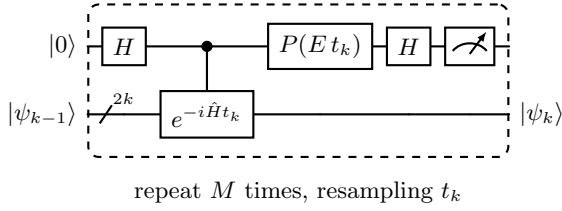


FIG. 10. One cycle of the rodeo algorithm: Hadamard on the ancilla, controlled time evolution $e^{-i\hat{H}t_k}$ of the $2k$ -qubit system register with a Gaussian-random time t_k , ancilla phase $P(Et_k)$, second Hadamard, and a mid-circuit measurement of the ancilla, which is then reset and reused for the next cycle. A run is accepted when all M outcomes read 0; each accepted cycle applies the filter $\frac{1}{2}(\mathbb{I} + e^{iEt_k}e^{-i\hat{H}t_k})$ to the system. Unlike the VQE of Fig. 9(a), the circuit carries no variational parameters and no classical optimization loop; the controlled evolution, compiled from the same Pauli-rotation blocks as Fig. 9(b) with only the central R_z controlled, is the sole system-size-dependent element ($\approx 4.2 \times 10^2$ CNOTs per Trotter step at $k = 4$).

CNOTs uncontrolled — 296 at $k = 4$ — and $\approx 4.2 \times 10^2$ CNOTs per step once the controls are added. A rodeo cycle with n_T Trotter steps therefore costs $\approx 4.2 \times 10^2 n_T$ CNOTs at $k = 4$.

B. The Trotterized pipeline

The results of Sec. IV E used exact sector exponentials; on hardware both the refinement slices and the controlled evolution are product formulas, and their error must be quantified at the circuit level. We split the Hamiltonian

exactly as the gate compilation does: a diagonal term D (kinetic energy plus the $p=q$ pairing term; single-qubit phases and ZZ rotations) and one pair-hopping term h_{pq} per level pair. The eight Pauli strings of each h_{pq} commute, so every factor $e^{-ih_{pq}\delta t}$ is realized *exactly* by its 48-CNOT block, and the only Trotter error is the non-commutativity between terms — exactly as on a device. At $k = 4$, $t = 1$ the second-order (symmetric) formula reaches spectral-norm error 3.4×10^{-2} with 4 steps and 8.3×10^{-3} with 8.

Figure 11 shows the two pipeline stages under this compilation. For the refinement (left), $n_s = 80$ second-order steps along the $T = 30$ path reproduce the exact-slice result to within its own adiabatic error, delivering $p = 0.9977$ against the exact-slice 0.9988; first-order steps need several hundred. For the rodeo stage (right) the honest scheme fixes the Trotter step *size* δt , so a cycle with Gaussian time t_k uses $\lceil t_k/\delta t \rceil$ steps ($\sigma\sqrt{2/\pi}/\delta t \approx 13$ on average at $\sigma = 4$, $\delta t = 0.25$). The filter is strikingly diagnostic of its own Trotter error: at $\delta t = 1$ the corrupted evolution is simply *rejected* — the acceptance collapses to 0.10 and the fidelity never converges — while at $\delta t = 0.25$ the infidelity tracks the exact-evolution curve down to 1.2×10^{-4} at $M = 6$ with acceptance 0.989. The intermediate $\delta t = 0.5$ converges to a floor of 2×10^{-3} : the rodeo filter projects onto the ground state of the *effective* Trotterized Hamiltonian, whose $O(\delta t^2)$ rotation away from the true ground state sets the attainable fidelity, confirming the self-correcting picture of Ref. [6] quantitatively.

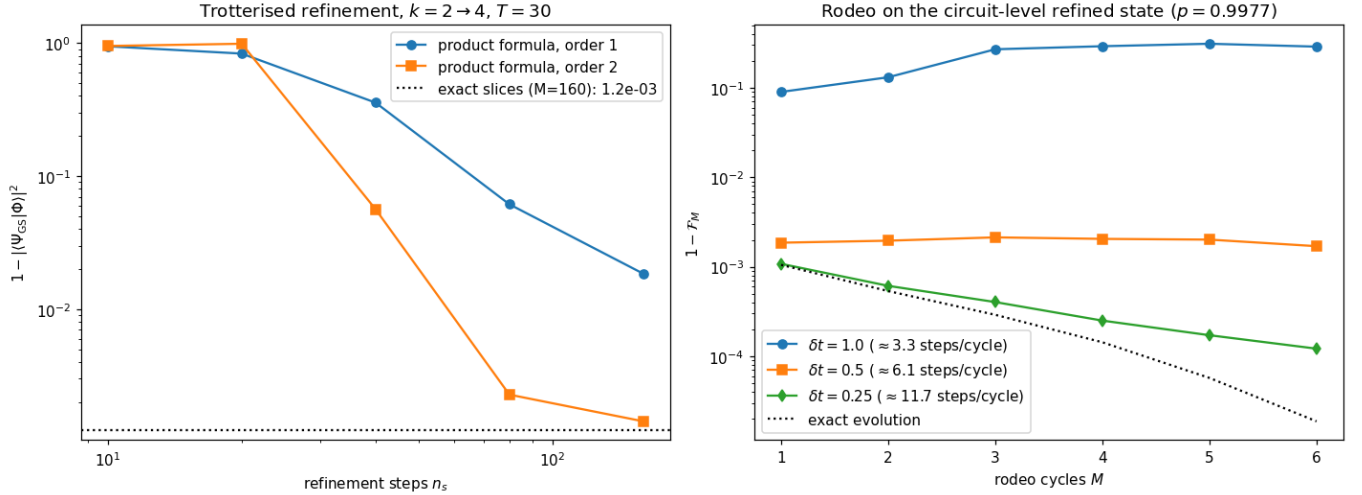


FIG. 11. The pipeline at the circuit level ($k = 4, N = 4, g = 1$). Left: infidelity of the Trotterized refinement versus the number of product-formula steps n_s along the $T = 30$ path, for first- and second-order formulas; the dotted line is the exact-slice reference. Right: rodeo infidelity versus the number of cycles M for a Trotterized controlled evolution with fixed step size δt (input: the $n_s = 80$ circuit-level refined state, $p = 0.9977$). Too-coarse steps are rejected by the filter itself (the $\delta t = 1$ acceptance collapses to 0.10); $\delta t = 0.25$ tracks the exact evolution.

C. Depth budget: UCCSD-VQE versus the rodeo-terminated pipeline

Table II summarizes the comparison at the circuit-level operating point of Fig. 11 ($n_s = 80$ second-order refinement steps, $M = 2$ rodeo cycles at $\delta t = 0.25$), counting *total CNOTs per accepted preparation at matched target*, including the post-selection repetition factor $1/P_M$. The comparison is deliberately honest in both directions.

The traditional route prepares the $k = 4$ ground state directly with the gate-level UCCSD-VQE: after a one-time classical optimization (eight L-BFGS iterations, each executing the full circuit once per energy and gradient evaluation, averaged over many shots), every subsequent preparation replays the 1312-CNOT, 26-parameter circuit. Its accuracy saturates at the ansatz error, $E - E_{\text{FCI}} = 1.4 \times 10^{-3}$ ($k = 4$, Table I), and the cost grows steeply with resolution ($272 \rightarrow 1312$ CNOTs from $k = 3$ to $k = 4$).

One pipeline repetition costs $80 \times 296 \times 2 = 47360$ CNOTs of parameter-free refinement plus ≈ 23000 CNOTs for two rodeo cycles (≈ 28 controlled Trotter steps of 416 CNOTs, doubled for the symmetric formula), i.e. $\approx 7.1 \times 10^4$ CNOTs per run; with acceptance $P_M = 0.99$ this is also the cost per *accepted* preparation. Per shot the pipeline is therefore ~ 50 times deeper than a single UCCSD replay — there is no free lunch in coherent depth, and the refinement dominates. What the depth buys is accuracy and trainability: the $M = 2$ operating point reaches an infidelity of 5×10^{-4} , four more cycles reach 1.2×10^{-4} , and with exact controlled evolution the same input reaches 5×10^{-6} (Fig. 8) — all below the variational floor, which no number of UCCSD-VQE shots can cross. The pipeline contains no variational

TABLE II. Circuit-depth budget at $k = 4$ ($N = 4, g = 1$, eight system qubits) at the operating point of Fig. 11: $n_s = 80$ second-order refinement steps ($T = 30$), $M = 2$ rodeo cycles with $\delta t = 0.25$, $\sigma = 4$. CNOT counts follow Table I and Eq. (8); the factor 2 is the symmetric-formula doubling. The UCCSD-VQE floor is its ansatz error; the pipeline numbers are per accepted preparation (acceptance $P_M = 0.99$).

circuit	params	CNOTs	infidelity
UCCSD-VQE prep. ($k=4$)	26	1312/prep.	1.4×10^{-3} (floor)
coarse UCCSD ($k=2$)	0	0	
prolongation $k=2 \rightarrow 4$	0	0	
refinement ($80 \times 296 \times 2$)	0	47360	2.3×10^{-3} (input)
rodeo, $M=2$ ($\approx 28 \times 416 \times 2$)	0	≈ 23000	5×10^{-4}
pipeline total per accepted prep.	0	$\approx 7.1 \times 10^4$	5×10^{-4}
— with $M=6$ cycles	0	$\approx 1.2 \times 10^5$	1.2×10^{-4}

parameters, so nothing is trained at full resolution; its dominant cost, the refinement, is set by $T^* \sim 1/\Delta E_{\text{min}}$ (Sec. IV D) and can be halved by starting the chain at $k = 3$ (Sec. IV C).

D. Remarks on adaptive and improved VQE

The steep UCCSD gate count is not irreducible, and much recent work aims precisely at shallower variational circuits (see Ref. [13] for a review). ADAPT-VQE [11] grows the ansatz one operator at a time, at each step appending the pool operator with the largest energy gradient and re-optimizing all amplitudes; it typically reaches a target accuracy with a small fraction of the

full UCCSD operator count and correspondingly shallower circuits. The pairing model makes the mechanism transparent: because the Hamiltonian conserves seniority and the reference is seniority-zero, the gradients of all singles and of all pair-breaking doubles vanish identically, so an adaptive selection would retain only the $\frac{N}{2}(k - \frac{N}{2})$ pair-hopping doubles — four operators at $k = 4$, $N = 4$, i.e. $4 \times 48 = 192$ CNOTs in place of the 1312 of the full 26-operator ansatz, a sevenfold compression, with the discarded amplitudes expected to vanish at the optimum just as the singles do. Qubit-ADAPT-VQE [12] pushes further by drawing the pool from individual Pauli strings, removing the eightfold string overhead per fermionic double. These developments are complementary to the present pipeline rather than competing with it: the pipeline is agnostic to how the coarse state is produced, so an ADAPT-style coarse solver slots in directly, reducing the (already modest) low-resolution cost and potentially raising the prolonged overlap; conversely, no variational compression by itself supplies the exponential, optimizer-free convergence that the rodeo filter provides once the overlap is large.

VI. DISCUSSION AND OUTLOOK

The pairing model lets us exhibit our entire hybrid pipeline with every stage benchmarked against exact results: a gate-level UCCSD coarse solver, a gate-free prolongation, a gapped adiabatic refinement whose $T^* \sim 1/\Delta E_{\min}$ scaling we verify directly through the shift-controlled path gap, and a rodeo filter with $1 - \mathcal{F}_M \lesssim \frac{1-p}{p} 2^{-M}$ — all validated a second time at the circuit level with explicit product formulas. The two algorithms are genuinely complementary — refinement provides the large overlap $p \approx 1$ that the rodeo algorithm requires, and the rodeo algorithm removes the slow adiabatic tail that refinement alone leaves behind. The coupling scan adds a practical lesson: at strong pairing the limiter is not the path gap but the quality of the prolonged starting state, which is exactly what starting the chain higher in the resolution hierarchy repairs.

Moving to hardware, the Trotterized pipeline of Sec. VB already quantifies the dominant coherent error: the rodeo algorithm filters the eigenstates of the effective (Trotterized) Hamiltonian [6], and its self-correcting design (small unitary errors broaden the energy window rather than biasing the estimate) keeps the required product-formula depth modest. The device-level requirements are then explicit: $2k + 1$ qubits (9 at $k = 4$, 17 at $k = 8$), mid-circuit measurement with ancilla reset and reuse, and — at the full $k = 4$ operating point of Table II — a coherent budget of order 10^4 – 10^5 two-qubit gates per accepted preparation.

a. The present hardware landscape. It is instructive to hold this budget against the machines available today. Among superconducting platforms, IBM’s Heron and Nighthawk processors (133–156 and 120 qubits, re-

spectively) reach median two-qubit errors at or below the 10^{-3} level and support dynamic circuits — mid-circuit measurement with feed-forward — with a stated target of executing ~ 5000 (growing toward 10^4) reliable two-qubit gates [39]; Rigetti’s 84-qubit Ankaa-3 reports 99.5% median two-qubit fidelity, with a chiplet-based roadmap toward 150-plus qubits at 99.7% [40]; and Google has demonstrated surface-code error correction below threshold [41], the gateway to logical depth on this platform. Trapped-ion systems trade speed for quality and connectivity: Quantinuum’s QCCD machines [42], most recently the 98-qubit Helios with 99.92% two-qubit fidelity, offer all-to-all connectivity and native mid-circuit measurement with qubit reuse — the primitive our rodeo cycle is built on — and the rodeo algorithm itself has already run on a Quantinuum H1-2 device in two-qubit form [31], as has its original demonstration on IBM hardware [6]. Neutral-atom arrays have moved fastest on logical qubits: reconfigurable atom arrays have run algorithms on 48 logical qubits with mid-circuit readout [43], and the Atom Computing/Microsoft stack has entangled 24 logical qubits and demonstrated logical computation [44] — the technology behind the Magne system of the QuNorth initiative in Copenhagen, specified at ~ 50 logical qubits on more than 1200 physical qubits [38].

For our pipeline the platform arithmetic is straightforward. A truncated demonstration — the $k = 3$ chain of Sec. IV C with a short refinement schedule and one or two rodeo cycles at coarse δt , roughly 10^3 – 10^4 two-qubit gates on seven qubits — is within reach of error-mitigated Heron/Ankaa-class superconducting devices and comfortable for Helios-class trapped-ion machines, whose all-to-all connectivity also absorbs the Jordan–Wigner ladders without SWAP overhead (on square or heavy-hex superconducting lattices the weight-4 pair-hopping rotations act on adjacent qubits, which limits, but does not eliminate, routing costs). The full $k = 4$ budget of $\sim 7 \times 10^4$ two-qubit gates per accepted preparation, by contrast, exceeds any physical-qubit error budget available today: at a two-qubit error of 8×10^{-4} the expected number of faults per run is $\mathcal{O}(50)$. It is, however, a natural early workload for the first logical-qubit generation: 9–17 logical qubits with deep coherent circuits and repeated mid-circuit measurement of a single logical ancilla is precisely the regime that Magne-class machines are being built for, and well below their specified logical-qubit counts.

b. Concluding remarks and perspectives. The present results establish, on a fully controlled example, that the hybrid of resolution refinement and the rodeo algorithm prepares correlated eigenstates with exponential accuracy at a cost set by an overlap that the refinement itself makes large — and they do so at the circuit level, with every stage compiled, budgeted, and validated against exact results. This work is deliberately the theoretical half of a two-step program: we are now moving from these considerations to calculations on existing and forthcoming machines. The staging follows

the platform arithmetic above. First, a mitigated $k = 3$ mini-pipeline (UCCSD coarse state, gate-free prolongation, short refinement, one to two rodeo cycles) on present-day trapped-ion or superconducting hardware, where the acceptance statistics of the rodeo filter — which, as Sec. VB shows, collapse when the compiled evolution is wrong — double as a sensitive hardware diagnostic. Second, the full $k = 4$ pipeline on an early logical-qubit machine such as Magne, for which the depth budget of Table II is a concrete workload specification. Third, scaling in k at fixed N (13 qubits at $k = 6$, 17 at $k = 8$, Table I) and onward to the realistic targets of this program — light and medium-mass nuclei benchmarked against coupled-cluster and in-medium similarity renormalization group calculations, molecular systems, and Hubbard models refined from coarse to fine lattices — with a closed-loop control layer that adapts the refinement schedule to the rodeo acceptance read out mid-circuit. The pairing model, exactly solvable at every resolution, will remain the calibration standard

against which those hardware campaigns are judged.

CODE AVAILABILITY

The complete implementation — pairing Hamiltonian, gate-level UCCSD-VQE, prolongation, adiabatic refinement, and the ancilla-based rodeo algorithm — is provided as a single Jupyter notebook that reproduces every figure in this article. Give GitHub link when everything is ready.

ACKNOWLEDGMENTS

The work of C.G. and G.F.L. acknowledge funding from the European Union’s Horizon Europe research and innovation program under the Marie Skłodowska-Curie Grant Agreement No. 101126636. M.H.-J. is partially supported by U.S. National Science Foundation (NSF) Grants PHY-1404159 and PHY-2310020.

-
- [1] A. Peruzzo, J. McClean, P. Shadbolt, M.-H. Yung, X.-Q. Zhou, P. J. Love, A. Aspuru-Guzik, J. L. O’Brien, *A variational eigenvalue solver on a photonic quantum processor*, Nat. Commun. **5**, 4213 (2014).
 - [2] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, H. Neven, *Barren plateaus in quantum neural network training landscapes*, Nat. Commun. **9**, 4812 (2018).
 - [3] T. Albash, D. A. Lidar, *Adiabatic quantum computation*, Rev. Mod. Phys. **90**, 015002 (2018).
 - [4] D. S. Abrams, S. Lloyd, *Quantum algorithm providing exponential speed increase for finding eigenvalues and eigenvectors*, Phys. Rev. Lett. **83**, 5162 (1999).
 - [5] K. Choi, D. Lee, J. Bonitati, Z. Qian, J. Watkins, *Rodeo algorithm for quantum computing*, Phys. Rev. Lett. **127**, 040505 (2021).
 - [6] Z. Qian, J. Watkins, G. Given, J. Bonitati, K. Choi, D. Lee, *Demonstration of the rodeo algorithm on a quantum computer*, Eur. Phys. J. A **60**, 151 (2024).
 - [7] S. Bogner *et al.*, *Quantum state preparation with resolution refinement*, Phys. Lett. B **875**, 140363 (2026).
 - [8] D. Lee, *Quantum techniques for eigenvalue problems*, Eur. Phys. J. A **59**, 275 (2023).
 - [9] J. Lee, W. J. Huggins, M. Head-Gordon, K. B. Whaley, *Generalized unitary coupled cluster wavefunctions for quantum computation*, J. Chem. Theory Comput. **15**, 311 (2019).
 - [10] A. Kandala, A. Mezzacapo, K. Temme, M. Takita, M. Brink, J. M. Chow, J. M. Gambetta, *Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets*, Nature **549**, 242 (2017).
 - [11] H. R. Grimsley, S. E. Economou, E. Barnes, N. J. Mayhall, *An adaptive variational algorithm for exact molecular simulations on a quantum computer*, Nat. Commun. **10**, 3007 (2019).
 - [12] H. L. Tang, V. O. Shkolnikov, G. S. Barron, H. R. Grimsley, N. J. Mayhall, E. Barnes, S. E. Economou, *Qubit-ADAPT-VQE: an adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor*, PRX Quantum **2**, 020310 (2021).
 - [13] J. Tilly, H. Chen, S. Cao, D. Picozzi, K. Setia, Y. Li, E. Grant, L. Wossnig, I. Rungger, G. H. Booth, J. Tennyson, *The variational quantum eigensolver: a review of methods and best practices*, Phys. Rep. **986**, 1 (2022).
 - [14] H. F. Trotter, *On the product of semi-groups of operators*, Proc. Am. Math. Soc. **10**, 545 (1959).
 - [15] M. Suzuki, *Generalized Trotter’s formula and systematic approximants of exponential operators*, Commun. Math. Phys. **51**, 183 (1976).
 - [16] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, S. Zhu, *Theory of Trotter error with commutator scaling*, Phys. Rev. X **11**, 011020 (2021).
 - [17] M. Hjorth-Jensen, M. P. Lombardo, U. van Kolck (eds.), *An Advanced Course in Computational Nuclear Physics*, Lecture Notes in Physics **936**, Springer (2017).
 - [18] M. A. Nielsen, I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press (2010).
 - [19] R. W. Richardson, *A restricted class of exact eigenstates of the pairing-force Hamiltonian*, Phys. Lett. **3**, 277 (1963).
 - [20] R. W. Richardson, N. Sherman, *Exact eigenstates of the pairing-force Hamiltonian*, Nucl. Phys. **52**, 221 (1964).
 - [21] J. Dukelsky, S. Pittel, G. Sierra, *Colloquium: Exactly solvable Richardson-Gaudin models for many-body quantum systems*, Rev. Mod. Phys. **76**, 643 (2004).
 - [22] E. Ovrum, M. Hjorth-Jensen, *Quantum computation algorithm for many-body studies*, arXiv:0705.1928 (2007).
 - [23] M. Q. Hlatshwayo, Y. Zhang, H. Wibowo, R. LaRose, D. Lacroix, E. Litvinova, *Simulating excited states of the Lipkin model on a quantum computer*, Phys. Rev. C **106**, 024319 (2022).

- [24] I. Stetcu, A. Baroni, J. Carlson, *Variational approaches to constructing the many-body nuclear ground state for quantum computing*, Phys. Rev. C **105**, 064308 (2022).
- [25] I. Stetcu, A. Baroni, J. Carlson, *Projection algorithm for state preparation on quantum computers*, Phys. Rev. C **108**, L031306 (2023).
- [26] A. Pérez-Obiol, A. M. Romero, J. Menéndez, A. Rios, A. García-Sáez, B. Juliá-Díaz, *Nuclear shell-model simulation in digital quantum computers*, Sci. Rep. **13**, 12291 (2023).
- [27] C. Jiang, J. Pei, *Quantum computing of the pairing Hamiltonian at finite temperature*, Phys. Rev. C **107**, 044308 (2023).
- [28] J. Zhang, D. Lacroix, Y. Beaujeault-Taudière, *Neutron-proton pairing correlations described on quantum computers*, Phys. Rev. C **110**, 064320 (2024).
- [29] D. Lacroix, E. A. Ruiz Guzman, Y. Beaujeault-Taudière, et al., *Symmetry breaking and symmetry restoration for many-body problems treated on quantum computers*, arXiv:2310.17996 (2023).
- [30] T. D. Cohen, H. Oh, *Optimizing the rodeo projection algorithm*, Phys. Rev. A **108**, 032422 (2023).
- [31] M. Bee-Lindgren, Z. Qian, M. DeCross, N. C. Brown, C. N. Gilbreth, J. Watkins, X. Zhang, D. Lee, *Controlled gate networks: theory and application to eigenvalue estimation*, arXiv:2208.13557.
- [32] M. Patkowski, O. Ayyildiz, K. Hunt, N. Jansen, D. Lee, *Improved rodeo algorithm performance for spectral functions and state preparation*, arXiv:2602.05978 (2026).
- [33] L. Lin, Y. Tong, *Near-optimal ground state preparation*, Quantum **4**, 372 (2020).
- [34] Y. Dong, L. Lin, Y. Tong, *Ground-state preparation and energy estimation on early fault-tolerant quantum computers via quantum eigenvalue transformation of unitary matrices*, PRX Quantum **3**, 040305 (2022).
- [35] M. Larocca, S. Thanasilp, S. Wang, K. Sharma, J. Biamonte, P. J. Coles, L. Cincio, J. R. McClean, Z. Holmes, M. Cerezo, *Barren plateaus in variational quantum computing*, Nat. Rev. Phys. **7**, 174 (2025).
- [36] D. Beck et al., *Quantum information science and technology for nuclear physics. Input into U.S. long-range planning, 2023*, arXiv:2303.00113 (2023).
- [37] C. W. Bauer, Z. Davoudi, N. Klco, M. J. Savage, *Quantum simulation of fundamental particles and forces*, Nat. Rev. Phys. **5**, 420 (2023).
- [38] QuNorth, *Magne: a level-2 neutral-atom quantum computer*, <https://qunorth.com> (accessed July 2026).
- [39] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. van den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, A. Kandala, *Evidence for the utility of quantum computing before fault tolerance*, Nature **618**, 500 (2023).
- [40] Rigetti Computing, *Rigetti launches the 84-qubit Ankaa-3 system, achieving a 99.5% median two-qubit gate fidelity milestone*, press release (December 2024), <https://investors.rigetti.com>.
- [41] R. Acharya et al. (Google Quantum AI and Collaborators), *Quantum error correction below the surface code threshold*, Nature **638**, 920 (2025).
- [42] S. A. Moses et al., *A race-track trapped-ion quantum processor*, Phys. Rev. X **13**, 041052 (2023).
- [43] D. Bluvstein et al., *Logical quantum processor based on reconfigurable atom arrays*, Nature **626**, 58 (2024).
- [44] B. W. Reichardt et al., *Logical computation demonstrated with a neutral atom quantum processor*, arXiv:2411.11822 (2024).